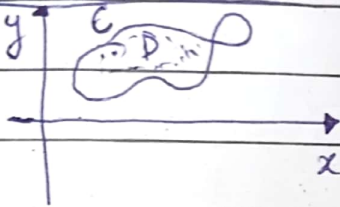


Overview of Line Integral & Single Integral.

$$\int_C F \cdot dx \quad \text{or} \quad \int_C G(x, y) \, ds$$

Basically Green's theorem gives a relationship b/w a line integral around a closed curve and a double integral over a region bounded by curve.

$$\boxed{\int_C P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA}$$



Exp:- Evaluate $\int_C x^4 \, dx + xy \, dy$, where C is the triangular consisting of the line segments from (0,0) to (1,0) from (1,0) to (0,1) and from (0,1) to (0,0).

Solution:-

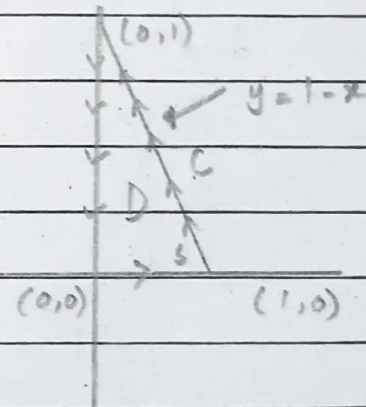
$$\int_C x^4 \, dx + xy \, dy$$

$$\int_C x^4 \, dx + xy \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$= \int_0^1 \int_0^{1-x} (y - 0) \, dy \, dx$$

$$= \int_0^1 \left[\frac{y^2}{2} \right]_0^{1-x} dx$$

$$= \int_0^1 \frac{1}{2} \left[(1-x)^2 - (0) \right] dx$$



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$$= \frac{1}{2} \int_0^1 (1-x)^2 dx$$

$$= \frac{1}{2} \int_0^1 (1 - 2x + x^2) dx$$

$$= \frac{1}{2} \left[1 - \frac{2x^2}{2} + \frac{x^3}{3} \right]_0^1$$

$$= \frac{1}{2} \left[1 - \cancel{2} \frac{(1)^2}{\cancel{2}} + \frac{(1)^3}{3} \right] - [0]$$

$$= \frac{1}{2} \left[\cancel{1} - \cancel{1} + \frac{1}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{3} \right]$$

$$= \boxed{\frac{1}{6}}$$